

COMMUNICATIONS

LOCALLY INDICABLE GROUPS OF FINITE MAL'TSEV RANK

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Recall that G is called a group of *finite Mal'tsev rank* r if every finitely generated subgroup of G is a group with r generators and r is the least number with this property. Below by the word 'rank' we mean the Mal'tsev rank. The notation and terminology adopted here are borrowed from [1, 2].

It is well known that the class of groups of finite rank is closed under subgroups, homomorphic images, and direct products of finitely many groups. For torsion-free Abelian groups, the standard concept of the rank of an Abelian group coincides with that of the Mal'tsev rank. We are also well aware that every torsion-free locally nilpotent group of finite rank is nilpotent and its nilpotency class does not exceed its rank. Groups of finite rank with additional restrictions were studied by many mathematicians such as S. N. Chernikov, V. S. Charin, and D. I. Zaitsev.

Research on groups of finite rank was reduced, as a rule, to imposing on a group conditions close to solvability. An impetus for writing this paper is the question whether every linearly orderable finitely generated group will be solvable. The structure of solvable matrix groups over $\mathbb{Q}$ is well known. By a theorem of A. I. Mal'tsev, these are finite extensions of triangular matrix groups over $\mathbb{Q}$. We know that every torsion-free solvable group of finite rank can be represented by matrices over the field of rational numbers [1]. Note also that if the rank of a torsion-free solvable matrix group over $\mathbb{Q}$ is finite then it is equal to the rational rank (D. I. Zaitsev).

In looking into the above question in detail, it became clear that its resolution reduces to treating a similar question for a wider class of groups, namely, the class of locally indicable groups. Recall that a group G is said to be *locally indicable* if every nontrivial finitely generated subgroup of G admits a homomorphism onto an infinite cyclic group. This class of groups has interesting group-theoretic properties. In [3], it was proved that every one-relator group having no elements of finite order is locally indicable.

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The class of locally indicable groups is closely connected with right orderable groups. In 1972 R. Burns and V. Hale proved that the class of locally indicable groups coincides with a class of groups admitting a right order under which the system of all convex subgroups of a group is a complete solvable system of subgroups (such groups are referred to as *Conradian*); see [1]. A well-known description of the structure of linearly orderable groups due to Mal'tsev implies that every linearly orderable group is Conradian. Also there is a close relationship between the class of linearly orderable groups and a wider class of Smirnov groups, i.e., right orderable groups whose systems of convex subgroups are solvable and infrainvariant [4].

An answer to the above question is the following:

THEOREM 1. Every finitely generated locally indicable group of finite rank is solvable.

An argument will be technically simpler if we consider a free group $F = F(x, y)$ with free basis x, y together an automorphism group Δ generated by standard automorphisms such as

$$\tau : x \rightarrow x^{-1}, y \rightarrow y, \quad \sigma : x \rightarrow x, y \rightarrow y^{-1}, \quad \zeta : x \rightarrow y, y \rightarrow x$$

(a dihedral group of order 8), and if a two-generated group G is treated as a factor group F/V , where $V^\Delta = V$. Such groups are said to be *symmetrically defined*.

The proof of the result mentioned reduces to examining two-generated locally indicable groups. In so doing, use is made of a refined version of the well-known theorem of B. Neumann on embedding finitely generated orderable groups in two-generated orderable groups (see [1]).

THEOREM 2. Every finitely generated right orderable group G is embedded in a two-generated right orderable group G^* . If G is Conradian, then G^* is also Conradian. If G has finite rank, then G^* has finite rank as well. If G is solvable of derived length s , then G^* likewise is solvable of derived length $s + 1$. If G is defined symmetrically and has finite rank, then G^* , too, is defined symmetrically and has finite rank.

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